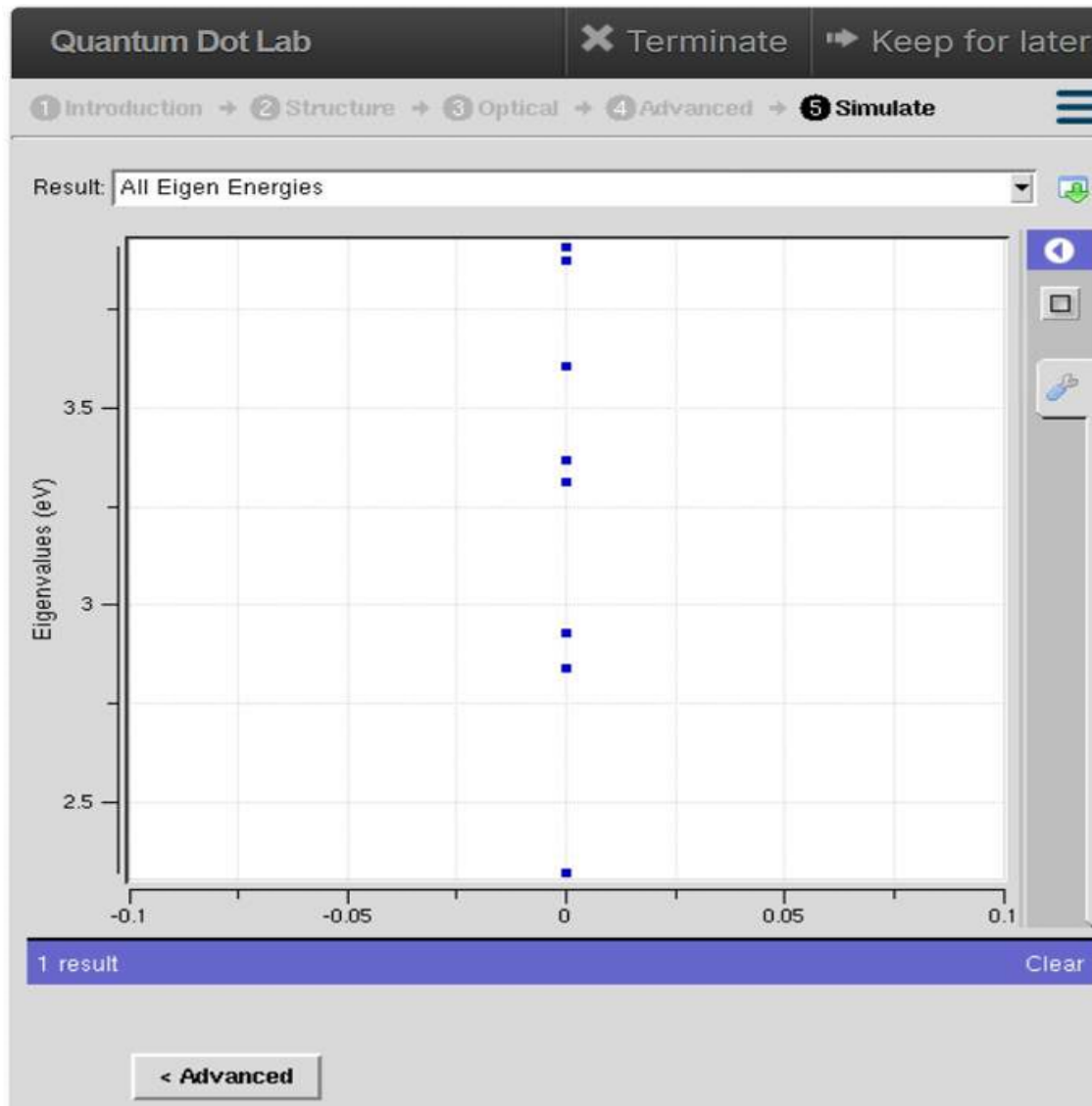
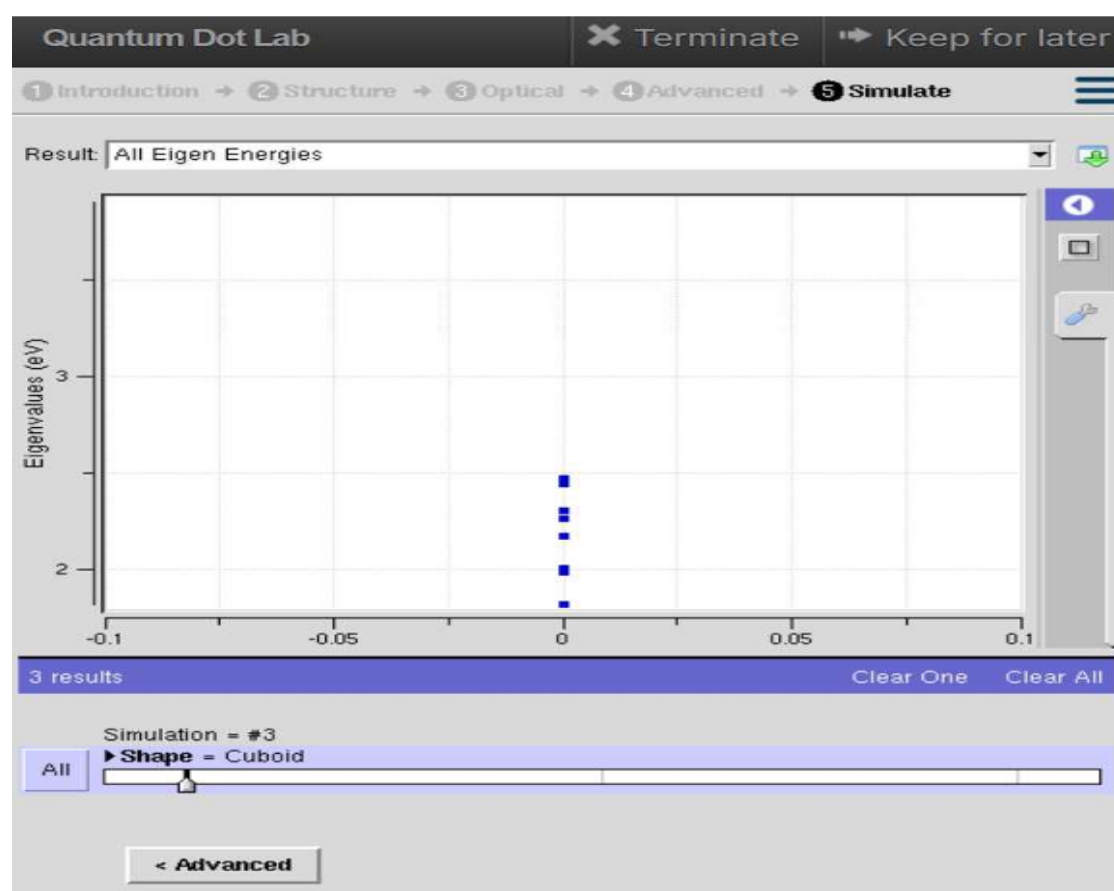


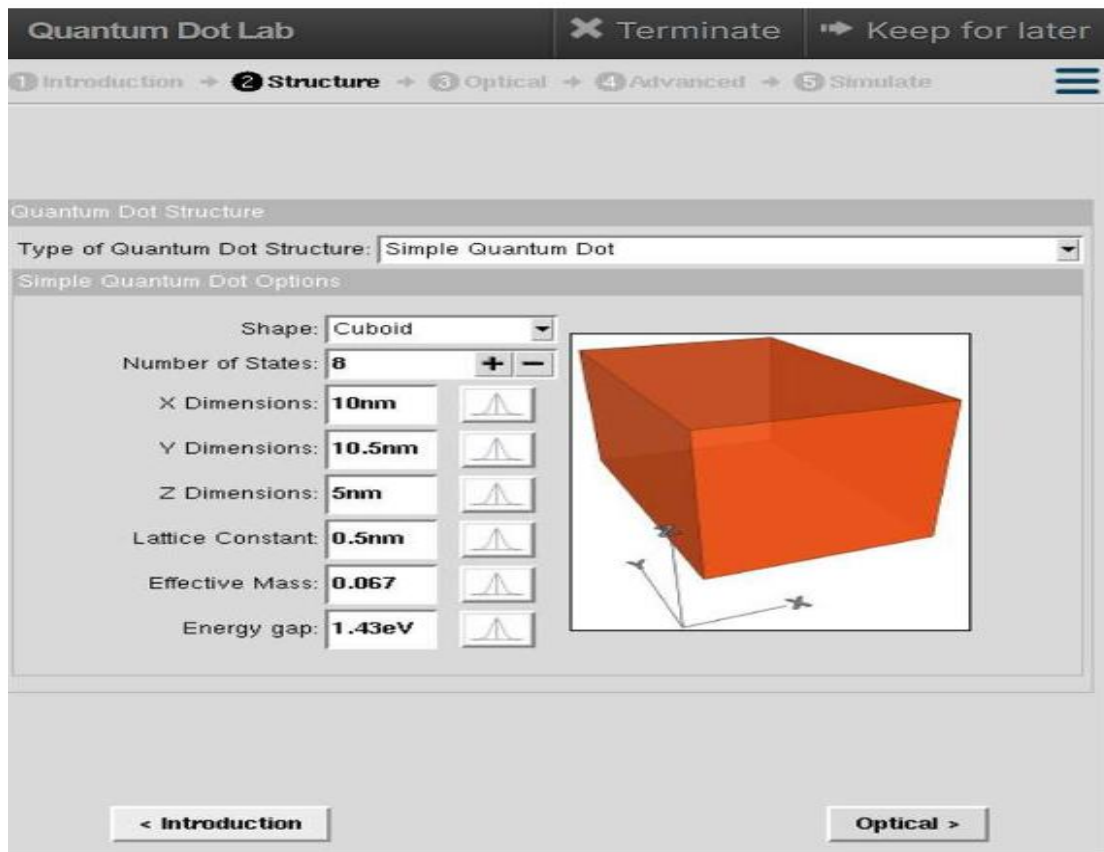
EXPERIMENT 3- SIMULATION OF ENERGY STATES OF QUAANTUM DOT USING NANO HUB



EXPLANATION:

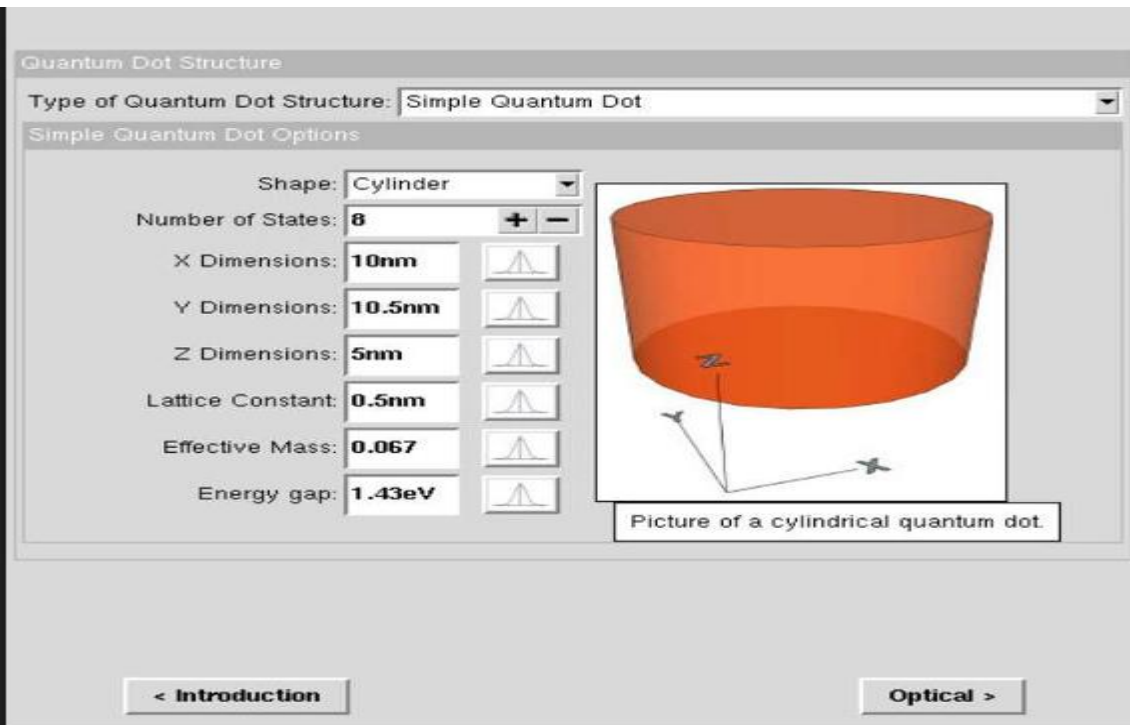
The simulation shows that the quantum dot has **discrete eigen energy levels** due to quantum confinement. These quantized energy states determine the optical and electronic properties of the quantum dot, making it useful for **LEDs, lasers, solar cells, and quantum computing applications.**





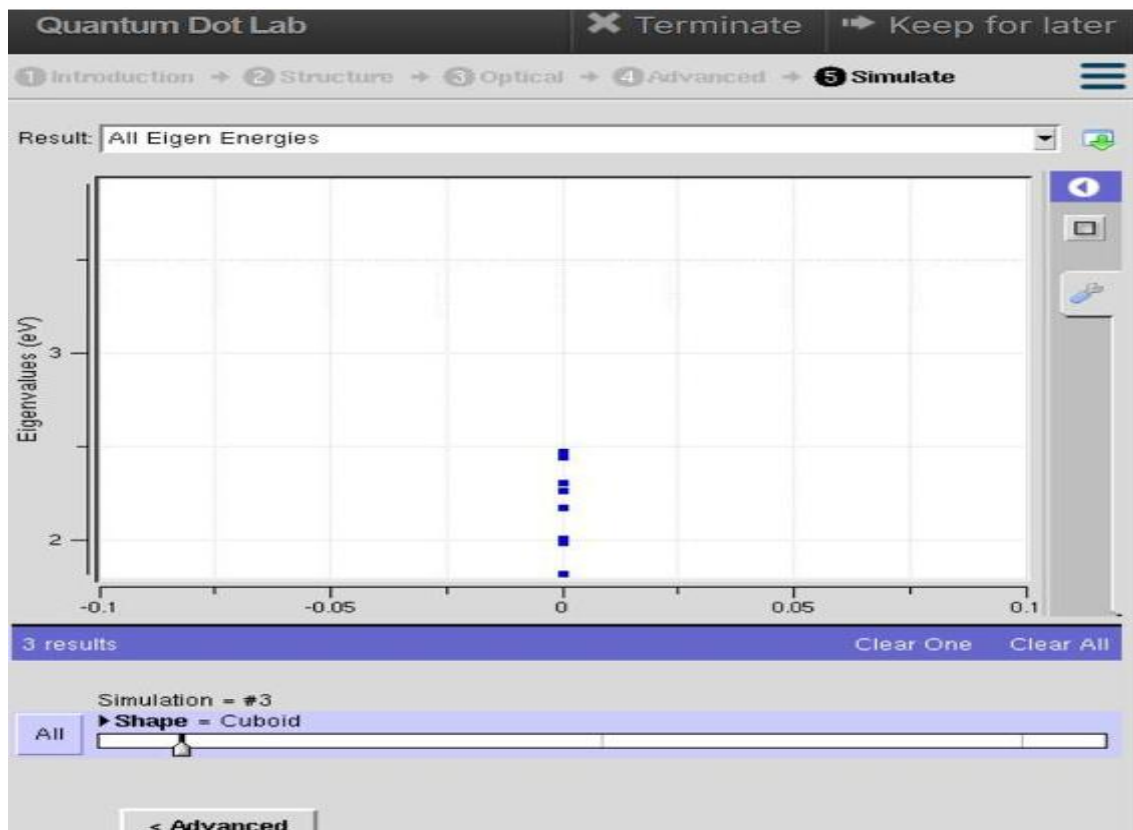
EXPLANATION:

The structure defines a **cuboid quantum dot** with specified dimensions and material properties. These parameters control electron confinement, resulting in **quantized energy levels** and influencing the electronic and optical behavior of the quantum dot.



EXPLANATION:

The simulation uses a **cylindrical quantum dot** with specified dimensions and material properties. The cylindrical confinement produces **quantized energy levels**, making the structure suitable for studying the electronic and optical behavior of quantum devices.



CONCLUSION:

The simulation of the cylindrical quantum dot demonstrates that electrons are confined in all three dimensions, resulting in discrete (quantized) energy levels. The size and shape of the quantum dot significantly influence its electronic and optical properties, making it suitable for nanoelectronic and optoelectronic applications.